Faculty of Information Technology
20241

ASPU
Al-Sham Private University

Multimedia-Lecture-Five

Audio

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Audio Signal

An audio signal is a representation of sound.

- It is typically using a level of electrical voltage for analog signals, and a series of binary numbers for digital signals.

- Audio signals have frequencies in the audio frequency range of roughly 20 to 20,000 Hz, which corresponds to the lower and upper limits of human hearing.

Dealing With Audio in Python

- Reading Audio File
- Playing Audio File
- Writing Audio File
- Find More Information about Audio
- How to Scaling Audio
- Drawing the audio signal

Reading Audio File:

```
>> data, samplerate = sf.read("test1.wav")
```

Import soundfile as sf library before read it

Writing Audio File:

```
>> sf.write(“output.wav”, data, samplerate)
```

Playing an audio:

```
>> os.system(output_file)
```

Find More Information About Audio:

```
>> info = sf.info(file_path)

>> print(f"Duration: {info.duration} seconds")

>> print(f"Sample rate: {info.samplerate} Hz")

>> print(f"Number of channels: {info.channels}")

>> print(f"Subtype: {info.subtype}")
```

Audio Format

Audio Scaling

Audio scaling allows for the modification of an audio signal amplitude or frequency.

Example1:

```
>> scaling_factor = 0.5
>> samplerate = samplerate * scaling_factor
```

Example2:

```
>> scaling_factor = 0.25
>> samplerate = samplerate * scaling_factor
```

Example3:

```
>> scaling_factor = 2.0
>> samplerate = samplerate * scaling_factor
```

Example4:

```
>> scaling_factor = 100
>> data  = data* scaling_factor
```

To increase the volume of the audio track you can multiple the variable it is stored in by a scalar.

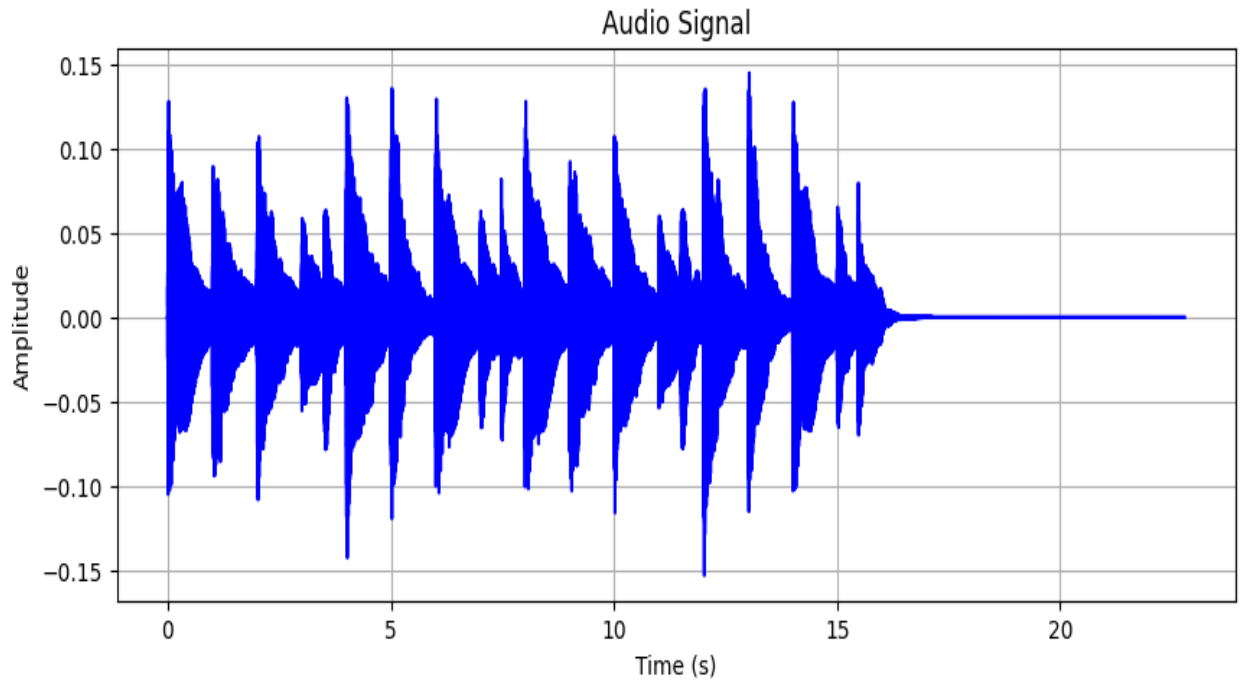
To slow down or speed up the track played you can adjust the sampling rate.

Drawing the audio signal

To draw an audio signal, you can easily use the function plot

Example:

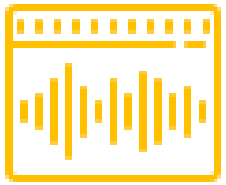
```
# Calculate the duration of the audio signal
duration = len(audio_signal) / sample_rate
# Create a time array
time = np.linspace(0, duration, len(audio_signal))
 # Plot the audio signal
plt.figure(figsize=(10, 4))
plt.plot(time, audio_signal, color='b')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.title('Audio Signal')
plt.grid(True)
plt.show()
```



Exercises:

I. Write a Python-code to:

a. Read an wav audio file.
b. Play the sound.
c. Reverse the signal using flipud function.
c. Reverse the two channels using fliplr function.
d. Save the new audio files on your disk.
e. Read the new audio files.
f. Play the two sound after flip.
g. plot three signal on one figure.



That’s All